Managing AI Like It Matters: The Artificial Intelligence Operating System (AI-OS)

Ewan Simpson

Narxoz Business School, Narxoz University

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Abstract

This paper presents the AI-OS, a governed and auditable operating system for human-AI work that treats AI as a tunable factor at the task layer. The AI-OS integrates four components into one decision-grade architecture. First, the AI Matrix (Access × Agency) provides a first-order principle that distinguishes empowerment from dependency across individual, task, job, and organizational levels. Second, the 3A Model supplies the strategic layer: Agency sets direction and decision rights, Alignment binds use to mission, values, and data responsibility, and Adaptation commits to short pilots, iterative learning, and vendor portability. Third, the Decision Compass operationalizes proportionality by scoring five factors to assign permitted modes of use (Autopilot, AI-Human Interface, Assist, Human Only) with prescribed controls, KPI templates, and appeal paths. Fourth, OI/IHACC provides the orchestration layer for knowledge work by designing end-to-end workflows across the data–information–knowledge–wisdom continuum while preserving human stewardship where stakes, ambiguity, or irreversibility are high. A task-first operating method links a live task register, residual human-effort estimation, role rebundling, and telemetry-based mode transitions. The paper advances testable propositions, specifies documentation and oversight practices that reduce opacity and automation drift, and outlines a research agenda on measurement validity, calibration, and field experiments. The contribution is a coherent system that improves reliability and role quality while keeping accountability with humans.

Keywords: AI governance; AI-OS; decision compass; human-AI collaboration; orchestration; AI transformation; organizational transformation.

1. Introduction

Organizations face a renewed version of an enduring design problem, namely how to allocate work between humans and machines in ways that raise performance without eroding judgment, fairness, or trust. The classic tradition showed that task decomposition and standardization can deliver scale in stable settings, although often at the cost of adaptability and intrinsic motivation (Taylor, 1911). Socio-technical systems research argued that technical and social subsystems must be designed together so that the introduction of tools is accompanied by redesigned roles, norms, and control systems (Trist & Emery, 1960). Contingency and information-processing perspectives added that there is no single best way, since structure should absorb the information load generated by uncertainty and interdependence through slack, lateral links, or decision rights (Galbraith, 1974). Work design research made the human consequences legible by identifying autonomy, skill variety, task identity, task significance, and feedback as primary levers of motivation and performance (Hackman & Oldham, 1976). These streams jointly suggest that the unit of analysis should be tasks, yet the design goal should be roles that concentrate human judgment and feedback while locating routine flow in scalable systems.

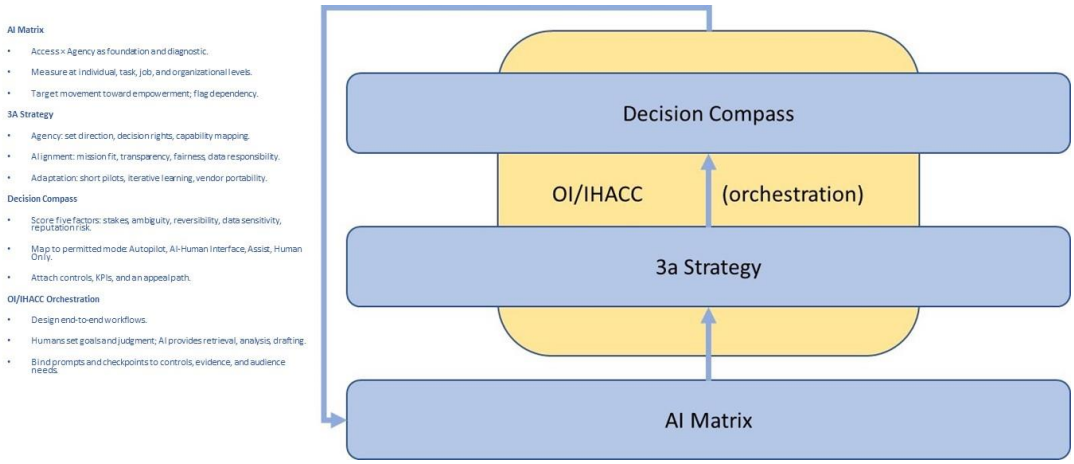
Artificial intelligence sharpens these debates. Modern systems can automate or assist perception, retrieval, drafting, and some forms of analysis, while limitations persist around ambiguous goals,

tacit norms, and high stakes judgment (Brynjolfsson et al., 2018). Task based economics indicates that technologies typically reallocate tasks rather than eliminate jobs, substituting for routine activities while complementing nonroutine and interpersonal ones, and sometimes inducing new human tasks through reinstatement effects (Autor et al., 2003; Acemoglu & Restrepo, 2018). At the same time, displacement risks are credible where high volume standardized processes dominate, which motivates proactive design for redeployment and role quality rather than passive attrition alone (Frey & Osborne, 2017). The techno social literature cautions that algorithmic management can shift discretion to opaque systems and create new forms of control, even when efficiency rises, which elevates the importance of oversight and appeal mechanisms (Kellogg et al., 2020).

AI also changes how knowledge is produced. The data–information–knowledge–wisdom architecture helps explain where AI contributes and where human stewardship remains essential. Systems accelerate transformation from data to information and often assist the move from information to knowledge, while the translation from knowledge to wisdom, which involves values, stakes, and trade-offs, remains a human and institutional responsibility (Ackoff, 1989; Rowley, 2007; Simpson, 2025a, 2025b). Research on cognitive offloading shows that externalizing memory or computation can increase capacity, although it may erode internal representation and vigilance if reflection and verification are weak (Risko & Gilbert, 2016; Gerlich, 2025). These epistemic concerns argue for designed 'human-in-the-loop' patterns that are proportionate to risk.

Organizations increasingly deploy AI inside everyday work, yet most efforts treat it as a set of tools rather than a governed system. The result is uneven performance, opaque decision paths, and fragile role design. This paper argues that reliable, defensible gains require an operating system that makes the human-AI boundary explicit, auditable, and adjustable by evidence. Figure 1 presents the Artificial Intelligence Operating System (AI-OS) as a layered architecture with a staged execution loop as discussed in detail in part 6 of the paper.

Figure 1. The AI-OS system



Source: Developed by the author.

In this paper we present the AI-OS as a governance-centered operating system for organizational AI. It integrates four components into one decision-grade architecture. First, the AI Matrix (Access × Agency) provides the first-order principle by turning access and agency into measurable targets and diagnostics across individual, task, job, and organizational levels. Secondly, we present the 3A

Model as the strategic layer: Agency sets direction and decision rights, Alignment binds use to mission, values, and data responsibility, and Adaptation commits to short pilots, iterative learning, and vendor portability. The Decision Compass is the operational layer that scores five factors (stakes, ambiguity, reversibility, data sensitivity, reputation risk) and assigns permitted modes with proportionate controls and appeal paths: Autopilot, AI-Human Interface, Assist, and Human Only. OI/IHACC is the orchestration layer for knowledge work, designing end-to-end workflows across the data–information–knowledge–wisdom continuum while preserving human stewardship where stakes, ambiguity, or irreversibility are high.

The AI-OS differs from ad hoc automation in three ways. First, it treats access and agency as managed variables and expects measurable movement toward empowerment rather than only hour savings. Second, it replaces tool-first decisions with risk-proportionate mode selection that is reversible when telemetry degrades. Third, it binds orchestration to governance so that prompts, checkpoints, and terminology are tied to documented controls, indicators, and appeal mechanisms. The operating method that follows instantiates this contract through a task register, residual-effort estimation, role rebundling, control and KPI libraries, and gated pilots that promote or revert modes by evidence.

This paper focuses on the conceptual and operational model. Illustrative deployments are omitted from the main text and are available separately under license. The contribution is an integrated system that consolidates first-order principles, strategy, operations, and orchestration into a coherent AI-OS that can be adopted as a transparent, auditable way to run human-AI work.

2. Literature Review

2.1 Classic foundations: decomposition, variance, and motivation

Organizational design has long wrestled with how to structure work for performance while preserving human capability and dignity. Scientific management demonstrated the power of task decomposition and standardization for throughput (Taylor, 1911), but proved brittle under uncertainty and corrosive to intrinsic motivation when tightly coupled to surveillance. Socio-technical systems (STS) reframed design as joint optimization of technical and social subsystems; technology must be introduced with parallel redesign of roles, norms, and control systems (Trist & Emery, 1960). Contingency and information-processing perspectives argued there is no single best way; structure should absorb the information load created by interdependence and uncertainty, via slack resources, lateral links, or richer decision rights (Galbraith, 1974). In parallel, work design research established that autonomy, skill variety, task identity, task significance, and feedback are primary levers of motivation and performance (Hackman & Oldham, 1976), implying that decomposition needs recombination into roles that sustain meaning and mastery. These streams jointly point to a modern principle: the unit of analysis for design should be tasks, but the design goal is roles that concentrate human judgment and feedback while locating routine flow in scalable systems.

2.2 Technological change as task reallocation

The classic routine versus nonroutine result still matters, but generative AI extends exposure into a wider set of cognitive tasks. A useful lens is task substitution, where AI takes over steps; task complementarity, where AI raises human throughput or quality; and reinstatement, where new human tasks emerge around the technology (Autor, Levy, & Murnane, 2003; Acemoglu &

Restrepo, 2019; Brynjolfsson, Mitchell, & Rock, 2018). Large displacement is not automatic. Outcomes depend on demand, reorganization costs, measurement, and governance.

For management, the decision is a portfolio of task-level choices. Early field studies show sizeable productivity gains from assistants, especially for less experienced workers, alongside heterogeneous effects that require proportionate controls. Reinstatement is visible in oversight, verification, exception handling, data curation, and prompt design. Mollick's co-intelligence framing is consistent with this: humans orchestrate goals and judgment while AI supplies scalable analysis, drafting, and search, yielding centaur or cyborg workflows rather than pure automation (Brynjolfsson, Li, & Raymond, 2025; Noy & Zhang, 2023; Acemoglu & Restrepo, 2019; Mollick, 2024). The operating question remains where substitution is safe, where complementarity yields the largest returns, and where reinstatement creates durable human value.

2.3 Techno-social risks: algorithmic management, opacity, and equity

Beyond reallocation, AI can reconfigure power and agency at work. Research on algorithmic management shows how datafied coordination and evaluation shift discretion from workers to opaque systems, generating new contestations even when efficiency rises (Kellogg, Valentine, & Christin, 2020). Risks include automation drift, explainability gaps, and inequity when data or models encode historical bias. Governance responses include internal audit frameworks and documentation practices that make decisions traceable and reviewable (Barocas & Selbst, 2016; Mitchell et al., 2019; Raji et al., 2020; Gebru et al., 2021). These concerns echo Science and Technology Studies and contingency theory: technologies are not neutral, and effects depend on local structures, discretion, and voice. Adoption should include appeal mechanisms, audit trails, escalation paths, and role-appropriate discretion, with attention to multi-level effects so that task efficiency gains do not erode autonomy at job or organizational levels (Barley, 1986; Orlikowski, 1992; Mateescu & Nguyen, 2019).

2.4 Knowledge production under AI: DIKW compression and epistemic risks

The DIKW hierarchy (the progression in complexity through Data; Information; Knowledge; Wisdom) remains a useful scaffold for locating where AI helps and where human judgment is essential. Contemporary systems accelerate movement from data to information and increasingly assist the information to knowledge step. Translating knowledge to wisdom remains a human and institutional task grounded in values, stakes, and trade-offs (Ackoff, 1989; Rowley, 2007). Simpson (2025a) argues that the development of AI merits a reconsideration of the paradigm.

Cognitive offloading research shows that externalizing memory or computation can raise capacity but can also erode internal representation and vigilance without strong controls and reflection. Generative AI extends offloading from memory to ideation and reasoning, which raises the bar for epistemic discipline through documented sources, counter-argumentation, and risk calibrated verification (Sparrow, Liu, & Wegner, 2011; Risko & Gilbert, 2016; Bender, Gebru, McMillan-Major, & Shmitchell, 2021; Gerlich, 2025).

2.5 Human-AI teaming: centaurs, co-intelligence, and human-in-the-loop

Managerial research increasingly frames AI as a teammate rather than an all-or-nothing substitute. "Centaur" patterns assign sub-tasks by comparative advantage in human-AI hybrids, and "co-intelligence" emphasizes orchestration: when to generate, retrieve, verify, or escalate

(Mollick, 2024; Kasparov, 2017). Human-computer interaction codifies this as human-in-the-loop design with interaction checkpoints, verification protocols, and explanatory interfaces calibrated to risk (Amershi et al., 2019; Holzinger, 2016). The consistent finding is that naive automation underperforms managed augmentation; the design object is the workflow, not the tool. This aligns with work-design and STS insights that effective teaming raises autonomy and feedback at the role level while reducing cognitive load on routine subtasks.

2.6 Orchestrated Intelligence (OI) and IHACC: choreographing hybrid knowledge work

OI/IHACC is the orchestration layer of AI-OS, turning prompts and checkpoints into decision-grade outputs without eroding stewardship. Orchestrated Intelligence (OI) reconceives the practitioner as an orchestrator who designs recursive, multi-audience workflows in which AI acts as a narrative-aligned contributor under human intent. Empirical demonstrations report time compression, consistency across formats, and translation across stakeholder contexts when prompts, checkpoints, and terminology are woven into a single causal logic (Simpson, 2025b). Iterative Human-AI Co-Creation (IHACC) provides the epistemic architecture: knowledge work advances through designed cycles across the DIKW continuum: framing and retrieval (data to information); composition and critique (information to knowledge); and audience-specific synthesis and decision (knowledge to wisdom) with humans binding outputs to purpose, audience, and ethics (Simpson, 2025a). OI/IHACC extends "centaur" and human-in-the-loop concepts from point interactions to end-to-end orchestration, making residual effort and oversight intensity first-class design parameters.

This extension connects directly to managerial and HCI guidance: co-intelligence emphasizes human orchestration of goals and judgment while AI supplies scalable analysis and drafting, and human-AI interaction guidelines formalize checkpoints, verification, and explanatory interfaces calibrated to risk (Mollick, 2024; Amershi et al., 2019).

2.7 Integrating the strands: from theory to a governed operating method

Synthesizing these literatures yields four design imperatives that ground the model advanced here:

Make the work visible at the task layer. Classic design and task-based economics require granularity to reason about interdependence, information load, and substitution or complementarity (Autor, Levy, & Murnane, 2003; Galbraith, 1974). Visibility is also a precondition for auditability and fair redistribution of effort.

Treat AI as a tunable factor with explicit guardrails**.** Use the 3A Model as the strategic frame (Agency, Alignment, Adaptation) and the Decision Compass to assign permitted modes by risk factors (Stakes, Ambiguity, Reversibility, Data Sensitivity, Reputation Risk). Modes are Autopilot**,** AI-Human Interface**,** Assist, with Human Only as the guardrail; each carries proportionate controls and verification (Amershi et al., 2019; Kellogg, Valentine, & Christin, 2020; Mollick, 2024).

Preserve and cultivate agency via the AI Matrix. The AI Matrix (Access × Agency) frames success as a portfolio shift toward empowerment across individual, task, job, and organizational levels, not merely hour savings. Identical tools yield divergent outcomes depending on access and discretion, which directs investment to raise both (Simpson, 2025c; Acemoglu & Restrepo, 2019).

Adopt an epistemic architecture for hybrid work**.** DIKW plus OI/IHACC justifies residual-effort estimation (what remains human and why), rebundling residual tasks into roles that maximize autonomy and feedback, and learning loops that update those estimates with telemetry while documenting sources, counter-arguments, and risk-calibrated checks (Risko & Gilbert, 2016; Rowley, 2007; Simpson, 2025a, 2025b; Mollick, 2024).

These imperatives define the AI-OS contract that links principles, strategy, operations, and orchestration.

2.8 Implications for displacement, role quality, and continuous redesign

Taken together, these strands justify moving beyond one-off automation toward a living capability for continuous redesign. First, displacement is addressed at the task level with transparent residuals and rebundling plans that elevate judgment, coaching, negotiation, and care, making role quality (autonomy, feedback) an explicit success metric alongside cost and cycle time (Hackman & Oldham, 1976; Noy & Zhang, 2023; Brynjolfsson, Li, & Raymond, 2025). Second, equity and due process require human-in-the-loop checkpoints proportionate to risk, appeal mechanisms, and explainable criteria in evaluation and deployment (Amershi et al., 2019; Kellogg et al., 2020; Raji et al., 2020). Third, strategy emphasizes dynamic capabilities: use telemetry on volumes, exceptions, and defects to sense opportunities, deliberately seize them by shifting the human-AI boundary with Compass modes under 3A guardrails, and transform by recomputing roles and headcount programmatically (March, 1991). Finally, adopting OI/IHACC practices generalizes benefits beyond operations to knowledge creation and translation, allowing organizations to climb DIKW faster without abandoning human stewardship at the apex (Simpson, 2025a, 2025b; Rowley, 2007; Mollick, 2024).

In sum, value is realized when organizations: (a) redesign who does what work with governed human-machine mode choices; (b) measure what remains human and why; (c) rebundle residual tasks into richer roles; (d) preserve agency and equity with proportionate oversight; and (e) run the whole system as a transparent, auditable product that improves with data. The remainder of the paper operationalizes this stance through the AI Matrix, the 3A Model and Decision Compass, and a task-first method from catalog to residuals to rebundling to headcount to pilot and scale.

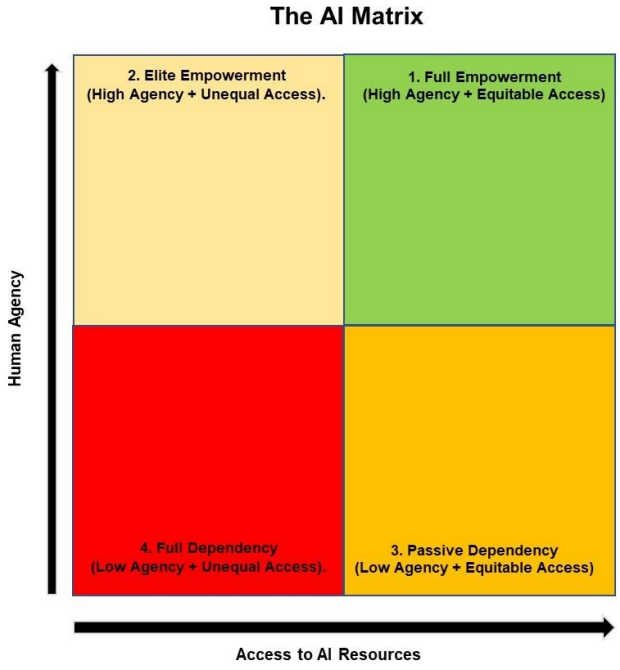
3. Conceptual Foundation: The AI Matrix as First-Order Principle

3.1 Access and Agency (the AI Matrix)

Within the AI-OS system, the AI Matrix is the foundation layer and the portfolio diagnostic for movement toward empowerment. The AI Matrix formalizes the idea that the effects of artificial intelligence depend on two conditions: access and agency. Access refers to the availability and usability of enabling resources, including tools, models, data, compute, interfaces, and integration infrastructure that make AI participation technically and economically feasible. Access includes not only technical provisioning but also training resources, documentation, and workflow integration that lower adoption costs for end users. Agency refers to the capacity of human actors to shape outcomes when using AI. Agency comprises discretion, skill and literacy, decision rights, override authority, and the social permission to exercise judgment. Agency is therefore a social and organizational construct as much as a cognitive one, since it is conferred and constrained by role design, evaluation systems, and governance practices (Hackman & Oldham, 1976; Kellogg, Valentine, & Christin, 2020).

As Figure 2 illustrates, crossing access and agency yields four outcome quadrants that describe likely trajectories for work and performance (Simpson, 2025c).

Figure 2. The AI Matrix



Source: Simpson (2025c).

Full Empowerment arises when workers have both reliable access to AI and genuine agency to use it. AI removes routine load and complements judgment, while autonomy and feedback rise, conditions linked to higher motivation and effectiveness; recent field studies show large gains when humans orchestrate AI rather than substitute wholesale (Hackman & Oldham, 1976; Noy & Zhang, 2023; Brynjolfsson, Li, & Raymond, 2025).

Elite Empowerment occurs when access or agency is concentrated in a narrow set of roles. A small group captures outsized productivity gains and discretion, while others see little change in autonomy or learning. This pattern is consistent with findings on algorithmic management that document shifts in control and opacity without deliberate governance for transparency and voice (Kellogg, Valentine, & Christin, 2020).

Passive Dependency arises when access is broad but agency is minimal. Individuals become operators of opaque systems and follow recommendations without proportionate verification or appeal, a configuration that increases the risk of automation bias and inequity if documentation, auditing, and appeal paths are weak (Raji et al., 2020; Selbst et al., 2019).

Full Dependency appears when both access and agency are low. The organization underuses AI while also constraining human judgment through rigid processes that cannot adapt to variance. As a diagnostic and steering device, the matrix directs managers to co-invest in access and agency; expanding only one dimension tends to produce mixed or negative social outcomes even where throughput improves (ILO, 2023; Kellogg et al., 2020).

The AI Matrix also coheres with task-based economics and socio-technical theory: when access and agency rise together, AI substitutes routine subtasks and complements nonroutine judgment, with reinstatement of new human tasks around governance, exception handling, and stakeholder translation (Autor, Levy, & Murnane, 2003; Acemoglu & Restrepo, 2019). When access rises without agency, adoption tends to amplify algorithmic control and opacity, degrading role quality despite local speed gains (Kellogg et al., 2020). Within AI-OS, the AI Matrix functions as both the foundation and the portfolio diagnostic for movement toward empowerment.

3.2 Multi-level framing

The AI Matrix is explicitly multi-level and tracks how empowerment or dependency can appear at the individual, task, job, and organizational levels, and how these levels interact over time.

Individual level. Access consists of usable tools, data permissions, and training; agency consists of decision rights, override authority, feedback loops, and evaluation criteria that reward appropriate judgment. Human-AI interaction guidance is clear: give users calibrated confidence, rationale capture, and recovery paths so that literacy and discretion rise together (Amershi et al., 2019).

Task level. The same technology can empower or disempower depending on design and reversibility. Proportionate controls and explainability at checkpoints keep automation from drifting beyond its remit. In practice, mapping stakes, ambiguity, reversibility, data sensitivity, and reputation risk to permitted modes with documentation improves traceability and due process (Amershi et al., 2019; Raji et al., 2020).

Job level. A job is a bundle of tasks. When routine subtasks move to AI and residuals are rebundled into roles with greater task identity, stakeholder proximity, and decision latitude, autonomy and feedback increase while coordination costs fall. Contingency and information-processing theory predict these differences; poorly matched structures amplify overload, well matched structures increase problem-solving capacity (Galbraith, 1974).

Organizational level. The matrix supports portfolio steering. Functions can be mapped by quadrant to see whether adoption is producing broad empowerment or narrow concentration. Leaders can then invest in decision rights, escalation protocols, and training where access is high and agency low, and in infrastructure where agency is high and access constrained. This aligns with dynamic capabilities and ambidexterity: sense misalignments with telemetry, seize by adjusting modes and rights, and transform by recomputing role architecture as conditions change (Teece *et al*., 2016; O'Reilly & Tushman, 2013).

Integration. Algorithmic management research shows how opacity can shift control away from workers even when task efficiency rises, while work-design links autonomy and feedback to engagement and quality; strategy research emphasizes adaptive structure under changing information loads. The AI Matrix links these by making access and agency explicit variables to measure and manage across levels (Kellogg et al., 2020; Hackman & Oldham, 1976; Galbraith, 1974).

3.3 Propositions

P1. Expanding access without expanding agency predictably increases dependency. When organizations scale access to models and data without enlarging discretion, override rights,

and literacy, control shifts toward opaque systems and away from accountable humans. At the task layer this appears as unexamined recommendations and weak appeal paths; at the job layer it downgrades roles into system operators. Evidence from algorithmic management shows how datafied coordination redistributes discretion unless governance, transparency, and voice are designed in. Work-design research links reduced autonomy to lower motivation and performance. Managers should therefore treat agency as a first-class design variable whenever access scales (Hackman & Oldham, 1976; Kellogg, Valentine, & Christin, 2020; Rosenblat & Stark, 2016; Raji et al., 2020; Selbst et al., 2019).

P2. Joint growth in access and agency improves role quality and adaptability. When access and agency rise together, AI substitutes routine subtasks and complements judgment, while remaining human work is rebundled into roles with greater task identity, autonomy, and feedback. Field evidence shows large productivity gains from generative assistants, especially for less-experienced workers, alongside heterogeneous effects that require proportionate controls; global exposure studies indicate augmentation dominates where governance and skills are in place (Noy & Zhang, 2023; Brynjolfsson, Li, & Raymond, 2025; ILO, 2023, 2025). This is consistent with co-intelligence and human-in-the-loop guidance that performance gains hinge on orchestration checkpoints, verification, and explainability calibrated to risk (Mollick, 2024; Amershi et al., 2019).

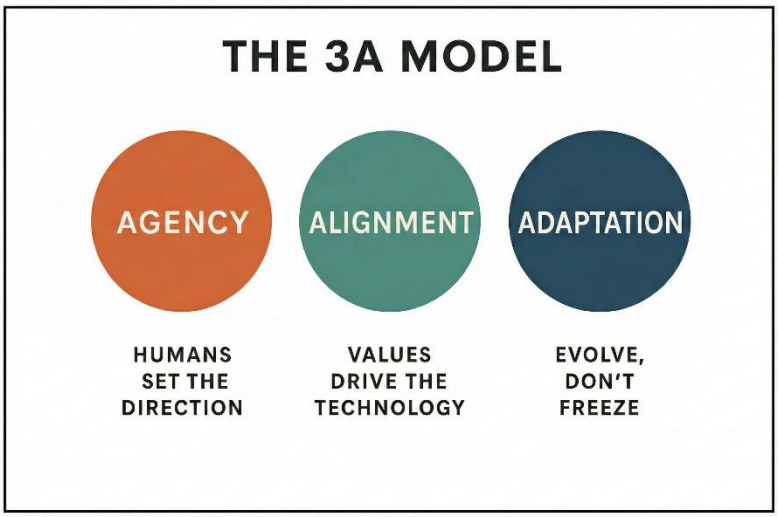
P3. Portfolio steering moves the enterprise distribution toward empowerment over time. If leaders measure access and agency by function and role, and deliberately invest to correct imbalances, the work portfolio shifts toward empowerment. The mechanism is iterative: sense misalignments through telemetry, seize opportunities by adjusting mode assignments and decision rights, and transform by rebundling tasks into coherent roles. This dynamic matches contemporary formulations of dynamic capabilities and organizational ambidexterity, which emphasize continuous recombination under uncertainty rather than one-off "automation initiatives" (Teece *et al*., 2016; O'Reilly & Tushman, 2013).

Together, these propositions establish the AI Matrix as a first-order design principle. The matrix supplies a normative target, i.e. movement toward empowerment through balanced investment in access and agency, and a diagnostic, i.e. a portfolio map that reveals where adoption is creating unintended dependency. The next sections operationalize this within the AI-OS using the 3A strategic layer, the Decision Compass, and the task-first operating method

4. The 3A Model: Internal Strategic Context for Human-AI Collaboration

In AI-OS terms, the 3A Model is the strategic layer that makes mode decisions and orchestration visible and reversible. The 3A Model summarized in Figure 3 sets the internal strategic context for the entire operating method.

Figure 3. The 3A Model



Source: Developed by the author

The 3A Model is concerned with why the organization is adopting AI and how it will govern the human-AI relationship over time. The model comprises three principles: Agency, Alignment, and Adaptation. These principles establish decision rights, ethical boundaries, and learning loops, and they provide a coherent bridge from values and strategy to the task-level assessments that follow. In this sense, the 3A Model is a governance and learning lens rather than a task-mode taxonomy.

4.1 Agency: Humans Set the Direction

Agency asserts that humans define institutional purpose, ethical boundaries, and acceptable trade-offs before selecting tools. Leadership articulates goals and constraints, then commissions capability mapping to ensure that workers and functions move from AI dependence toward empowerment. Practically, capability mapping is carried out through the Decision Compass and portfolio views such as the Agency–Engagement Matrix, which together identify where access and agency must be increased to produce better outcomes. This principle is consistent with socio-technical systems theory, which requires joint optimization of technical and social subsystems, and with work design research that links autonomy and feedback to sustained performance (Hackman & Oldham, 1976; Trist & Emery, 1960). In operational terms, AI is deployed both to automate where standards are clear and to amplify human creativity, critical thinking, and problem solving where judgment is central, reflecting task-based economics on substitution and complementarity (Acemoglu & Restrepo, 2018; Autor, Levy, & Murnane, 2003).

4.2 Alignment: Values Drive the Technology

Alignment requires that AI use be tested against organizational mission and strategy and that ethical guardrails ensure transparency, fairness, and data responsibility. In practice, this means that mode assignments and control choices must be explainable and auditable, that data sensitivity is considered in advance, and that due process is protected through appeal mechanisms. These expectations parallel guidance from human-AI interaction research, which emphasizes visible system status, control affordances, and verifiable handoffs, as well as the algorithmic management literature, which cautions against opacity and uncontrolled shifts in discretion (Amershi et al., 2019; Kellogg, Valentine, & Christin, 2020). Alignment operationalizes

the normative core of the AI Matrix by making social outcomes such as empowerment and role quality explicit management objectives, not by-products.

4.3 Adaptation: Evolve, Do Not Freeze

Adaptation commits the organization to iterative learning. Implementation begins with low-risk, high-impact pilots and progresses through short learning cycles with predefined acceptance criteria. The distribution of tasks across modes is revised by evidence, not assumption, and the organization maintains vendor independence through data portability, model documentation, and multi-platform strategies. Adaptation expresses a dynamic capabilities posture: sensing performance and risk through telemetry, seizing improvement opportunities through controlled mode transitions, and transforming role architecture and staffing as lessons accumulate (Teece *et al*., 2016). The principle is also consistent with contingency and information-processing perspectives that call for structures which absorb changing information loads rather than locking in a single design solution (Galbraith, 1974).

4.4 Implications for the Operating Method

The 3A principles condition every AI-OS step. Agency turns capability mapping into a first-class activity, which in our method occurs through the Decision Compass and task register discussed below. Alignment requires that each task have explicit controls and key performance indicators, and that the human-AI boundary be explainable to stakeholders. Adaptation requires telemetry, review cadences, and reversible choices, so tasks can move toward greater automation when warranted or return to higher-human-control modes when risk or drift is detected. Together these principles ensure that the operating method remains ethically grounded, strategically coherent, and empirically revisable.

5. The Decision Compass in AI-OS: From risk scoring to mode selection and controls

In AI-OS, the Decision Compass is the operational layer that converts risk into permitted modes and controls. The Decision Compass operationalizes the 3A principles at the task layer. It scores each task on five factors and converts the profile into a recommendation about the permitted mode and the control package. The result is an evidence based and auditable way to decide when automation is appropriate and when decisions must remain under human control.

5.1 Purpose and scope

The Compass serves three purposes. First, it standardizes assessments across functions so that similar risks receive similar treatments. Second, it ensures proportionality by matching oversight strength to risk. Third, it guarantees explainability by recording factor scores, rationales, and control choices for review by workers, auditors, and external stakeholders. A trained agent and process owners co-administer the Compass, and outputs populate the task register that anchors portfolio steering and role rebundling. These design choices follow socio-technical guidance on joint optimization, information-processing fit, and human-AI interaction principles for clear handoffs and visible system status (Amershi et al., 2019; Galbraith, 1974; Trist & Emery, 1960).

5.2 Factors and scales

Each task is scored on five domain anchored factors:

- Stakes: consequence of errors or delay.
- Ambiguit**y**: degree of judgment, context, or contestable standards.
- Reversibility: ease of recovery from error.
- Data sensitivity: presence of personal, confidential, or regulated data.
- Reputation risk: likelihood of public or brand harm.

Anchored rubrics and calibration sessions improve interrater reliability. For consequential tasks, scoring authority is separated from operational ownership to protect integrity. Single factor ceilings can block lower oversight modes when, for example, stakes are high or reversibility is low.

5.3 Mapping scores to modes

Composite scores map to permitted modes:

- Autopilot
- AI-Human Interface
- Assist
- Human Only

This mapping encodes proportionality and supplies a transparent rule for boundary decisions. Organizations may impose guardrails that prevent Autopilot when any single factor exceeds a ceiling. Every assignments carries a change record so that promotions or reversions are traceable.

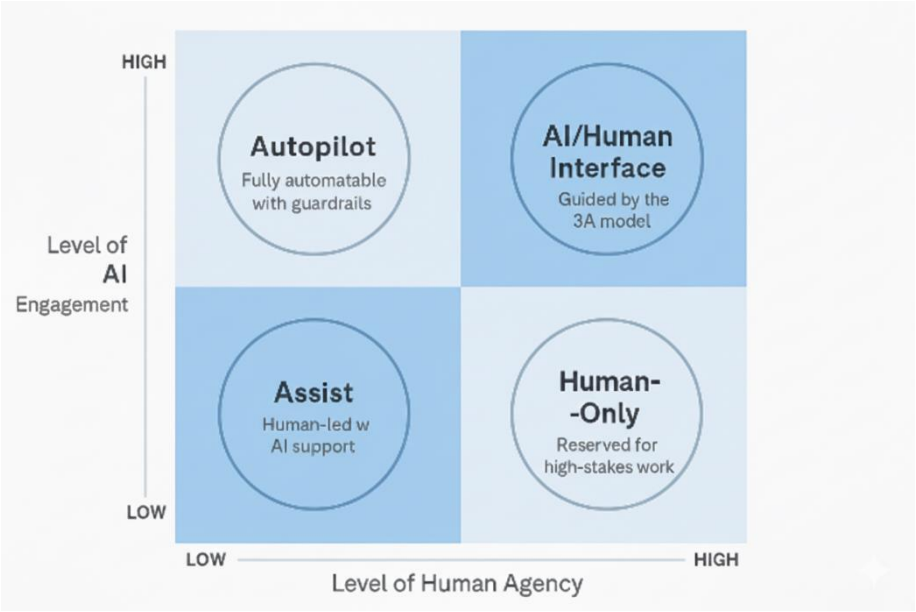
5.4 Mode definitions in the AI-OS Decision Compass context

Modes are specified here as the outcome of Compass scoring. Figure 4 illustrates how the modes fit with the AI Matrix.

- Autopilot authorizes automatic execution when inputs meet standards and controls are present. Typical controls include input validation, exception queues with service levels, periodic sampling, and drift monitoring. Expected benefits include reduced cycle time and manual hours with maintained or improved defect rates.
- AI-Human Interface standardizes interaction around system recommendations. The system surfaces options, evidence, and uncertainty; the human adjudicates within a decision frame. Controls include rationale capture, uncertainty display, change-of-decision logging, and dual sign-off where reversibility is limited. Expected benefits include improved consistency and explainability in variable contexts.
- Assist provides suggestions for retrieval, drafting, or analysis while preserving human ownership of the output. Controls include verification checklists, source citation or logging, and adversarial review for consequential decisions. Telemetry should distinguish acceptance, modification, and rejection of suggestions.
- Human Only reserves decision authority for accountable humans when stakes or ambiguity are high, reversibility is low, or equity and rights considerations dominate. AI may be used as advisory input, but the decision record reflects human reasons and due process. Controls include structured deliberation templates and peer review.

These definitions follow the proportionality logic of the Compass and the governance posture of the 3A Model. They also respond to cautions from the algorithmic management literature by keeping the human-AI boundary visible and contestable where it matters most (Kellogg, Valentine, & Christin, 2020).

Figure 4. Decision Compass modes mapped to the AI Matrix



Source: Developed by the author

5.5 Control libraries and KPI templates

For each mode, the Compass prescribes a primary control and a key performance indicator template:

- Autopilot**:** input validation, exception queues, periodic sampling; KPIs for defect rate and cycle time.
- AI-Human Interface: rationale capture, uncertainty display, override logging; KPIs for agreement rates, override frequency and reasons, and decision latency, with equity screens for people-affecting outputs.
- Assist**:** verification checklists and evidence citation; KPIs for post-verification accuracy, edit distance, and time to insight.
- Human Only: structured deliberation and peer review; KPIs for due process completion and stability across comparable cases.

These prescriptions align with human-AI interaction guidance that high risk or high ambiguity decisions require richer interfaces and stronger records, while low risk decisions can rely on sampling and aggregate monitoring (Amershi et al., 2019; Holzinger, 2016).

5.6 Documentation, appeals, and explainability

Every task entry captures factor scores, mode assignment, control package, KPI template, and escalation path. Appeals for consequential decisions reference factor rationales and the control record. Model documentation and data lineage artifacts strengthen traceability and internal audit, addressing well documented concerns about opacity and shifting control (Mitchell et al., 2019; Gebru et al., 2021; Raji et al., 2020; Kellogg et al., 2020).

5.7 Evidence, pilots, and mode transitions

Assignments are provisional and subject to evidence. Green light candidates proceed to short pilots with predefined acceptance criteria. Telemetry includes cycle time, defect or reject rates, exception and override frequencies, and equity screens for people-affecting outputs. Mode transitions are approved when performance remains within thresholds and risk is controlled, or reversed when drift or equity concerns arise. This learning loop implements the Adaptation principle and expresses a dynamic capabilities posture of sensing, seizing, and transforming (Teece *et al*, 2016).

5.8 Integration with the Agency-Engagement Matrix and task mapping

The AI-OS agent compiles Compass outputs to map tasks and functions to the Agency Engagement Matrix. This portfolio view situates tasks by empowerment trajectory, reveals concentrations of access without agency, and directs capability building and decision rights to move units toward empowerment quadrants. Large enterprises can map hundreds of tasks across dozens of functions, creating a transparent and auditable system that shows who does what, how AI is used, and where human judgment remains critical. This integration realizes the Agency principle at scale and creates the managerial surface for continuous redesign.

6. Operating method: the AI-OS execution loop

This section describes and specifies an operating method that renders the human-AI boundary visible, governable, and auditable. The unit of analysis is the task, and the design goal is the role. The method proceeds through ten stages that connect decomposition to rebundling and portfolio steering, while preserving traceability from assumptions to outcomes. Each stage produces defined artifacts, control choices, and measurement, which enables explanation to internal and external stakeholders and supports replication. These are summarized in Figure 5 with a brief commentary on each stage following.

Figure 5. Operating method stages

Stage	Objective	Outputs	Primary KPIs or Acceptance Criteria
6.1 Catalog the work at the task layer	Make work visible and auditable	Task register as traceability spine	Baseline cycle time; manual time; defect or reject rate
6.2 Assign provisional mode and estimate residual human effort	Make the human-AI boundary explicit	Mode tag per task; residual percentage per task	None prior to Compass; residuals serve as priors
6.3 Apply Decision Compass and attach controls and indicators	Ensure proportionality and explainability	Permitted mode; control package; acceptance metric	Accuracy or defect rate; timeliness; override frequency; edit distance; equity screen as applicable
6.4 Rebundle residual work into coherent roles	Convert residual tasks into accountable roles	Role catalog with responsibilities, rights, controls, indicators	Role-level measures of autonomy, feedback, task identity; exception service levels
6.5 Convert hours to headcount and surface governance exposure	Quantify capacity and identify risk concentration	FTE projections using internal rules; governance flags; consolidation backlog	Throughput vs. plan; audit trail for FTE computation; attribution of changes to pilots and transitions
6.6 Pilot, measure, and scale via portfolio steering	Institutionalize learning and prevent drift	Approved transitions; updated residuals; pilot reports	Stability of accuracy, timeliness, and equity within thresholds
6.7 Measurement, telemetry, and learning loops	Update priors and maintain validity	Quarterly review pack; updated parameters and thresholds	KPI deltas from baseline; reasons for overrides; appeal outcomes; role-quality scores
6.8 Data governance and documentation	Ensure compliance and reproducibility	Governance dossier per process	Compliance findings; documentation completeness
6.9 Boundary conditions and contingencies	Tune parameters to context	Context-specific parameter set	Domain-fit checks; exception volumes
6.10 Expected benefits	Link method to outcomes and next steps	Benefits summary; next-cycle roadmap	Task level: cycle time and defects; Role level: autonomy and feedback; Enterprise level: credible FTE and empowerment shift

6.1 Catalog the work at the task layer.

Make work visible and auditable by building a task register that becomes the traceability spine. Establish baselines for cycle time, manual time, and defect or reject rates. Where measurement is weak, record ranges and assumptions and plan instrumentation.

6.2 Assign provisional mode and estimate residual human effort.

Make the human-AI boundary explicit by tagging each task with a provisional mode and a residual percentage that serves as a prior. Do not treat these as final; defer contentious calls to Decision Compass scoring in Stage 3.

6.3 Apply Decision Compass and attach controls and indicators.

Ensure proportionality and explainability by scoring the five factors and setting the permitted mode, the primary control, and the acceptance metric. Track accuracy or defect rate, timeliness, override frequency, edit distance, and equity screens where people are affected. Apply single-factor ceilings when risk is high and separate scoring authority from operational ownership for consequential tasks.

6.4 Rebundle residual work into coherent roles.

Convert residual tasks into accountable roles with clear responsibilities, decision rights, controls, and indicators. Aim to increase task identity, autonomy, feedback, and stakeholder proximity. In low-volume settings, consider shared services to avoid fragmentation.

6.5 Convert hours to headcount and surface governance exposure.

Aggregate after-AI hours by role and produce FTE projections. Convert after-AI hours to headcount using the organization's standard capacity convention. Flag governance hotspots where Compass scores, overrides, or sensitive data concentrate. Standardize capacity reporting and record parameter changes across cycles.

6.6 Pilot, measure, and scale via portfolio steering.

Run short, gated pilots with predefined acceptance criteria and collect telemetry on stability of accuracy, timeliness, and equity. Promote, hold, or demote modes based on evidence and report Green-light coverage. Use conservative settings in safety-critical areas and require second-party review where reversibility is low.

6.7 Measurement, telemetry, and learning loops.

Update priors and thresholds using process telemetry, audits, surveys, and equity screens. Monitor cycle time, manual time, rejects, overrides, and appeal outcomes, and track due-process completion for Human Only decisions. Tighten controls or revert modes if signals degrade and keep all changes explainable.

6.8 Data governance and documentation.

Ensure compliance and reproducibility with a governance dossier per process that captures sources, permissible use, retention, model documentation, and decision records. Strengthen requirements in regulated domains and minimize exposure by design.

6.9 Boundary conditions and contingencies.

Tune thresholds, controls, and evidence bars to domain risk and reversibility profiles. Use peer review in creative or novel contexts and run domain-fit checks while monitoring exception volumes. Record justifications for parameter differences and revisit as data improve.

6.10 Expected benefits.

Link the method to outcomes at task, role, and enterprise levels and set next-cycle priorities. Report credible FTE effects using the organization's capacity conventions and movement toward

empowerment, alongside role-quality signals such as autonomy and feedback. Keep claims evidence-based and publish rubrics that support replication.

6.11 Summary

This section has laid out a governed workflow from inventory to impact: first, make work visible by cataloging tasks and baselining time, defects, and volumes; second, mark each task with a provisional mode and residual effort prior; third, use the Decision Compass to score risk, set the permitted mode, and attach proportionate controls and acceptance indicators. Residual tasks are then rebundled into coherent roles, after-AI hours are converted to FTE using the organization's standard capacity convention, and governance hotspots are flagged. Short, gated pilots collect telemetry on accuracy, timeliness, overrides, and equity, promoting or reverting modes by evidence. Continuous measurement updates priors and thresholds, while a documentation dossier keeps sources, decisions, and appeals traceable. Data governance and boundary settings are tuned to domain risk and reversibility, with Human Only guardrails where needed. The result is a transparent, auditable operating method that improves cycle time and reliability, raises role quality, and steers the portfolio toward empowerment. This loop closes the AI-OS contract by binding task-level mode choices to telemetry and portfolio steering on the AI Matrix.

7. Limitations and future research

7.1 Limitations

Any model has limitations. We have presented a coherent, grounded method for practical AI application for organizations, we acknowledge a range of limitations at the current stage of development. In this section these are laid out in the hope of stimulating further discussion.

Conceptual scope. The model is a governed operating grammar rather than a theory of adoption dynamics. It specifies how to design and audit human-AI work once an organization chooses to use AI, but it does not explain why firms adopt at a given time or intensity. The AI Matrix is a useful first-order lens, yet access and agency are composite constructs. Without validated measures, movement across quadrants can be misread as empowerment when it is only tool availability or as dependency when it is temporary learning cost.

Measurement and instrumentation. The Decision Compass relies on five factors scored by trained actors. Scoring remains susceptible to rater effects, strategic misreporting, and domain drift as practices evolve. Residual human effort is treated as an adjustable prior that is updated by telemetry, but the quality of telemetry varies across processes, and acceptance metrics can be gamed. The capacity conversion to FTEs based on internal standards is a convention rather than a law of nature. In weak data environments the resulting headcount projections carry wide error bands.

Mode misclassification risk. Composite scores can hide single factor hazards. Even with ceilings, tasks can be promoted to Autopilot before evidence supports stability. Interface and Assist can also create an illusion of oversight if rationale capture degrades into perfunctory compliance. Human Only guardrails help, but the organization must be willing to revert modes when defect rates, equity screens, or appeal patterns deteriorate.

Governance and culture. The model presumes that leadership is willing to make agency a design variable and that due process is valued. In low trust environments or where performance

management is punitive, documentation and appeals can be perceived as surveillance rather than protection. Participation by works councils or employee representatives may be required to secure legitimacy in some jurisdictions. Vendor portability and documentation requirements can add overhead that small teams struggle to carry.

External validity and heterogeneity. The conceptual illustration in this paper is deliberate. Sectoral differences in risk, reversibility, and regulation are large. Safety critical contexts, public services, and low resource settings require stricter ceilings, stronger documentation, and longer stabilization periods than office workflows. The portability of KPI templates, control libraries, and calibration guides should be tested across languages, legal regimes, and data infrastructures before general claims are made.

Causal inference and learning effects. Evidence cited for productivity gains often comes from short trials or early deployments. Learning curves, substitution effects, and organizational reconfiguration play out over longer horizons. Without careful counterfactuals and attention to sequencing, observed improvements can conflate tool novelty, selection effects, and concurrent process cleanups. Mode transitions triggered by noisy telemetry risk enshrining local optima.

7.2 Future research

At the enterprise level under AI-OS, test whether measured movement toward empowerment correlates with cycle-time stability, defect rates, and reliability. In parallel, develop and validate separate indices for access and agency. Develop and validate indices that separate tool availability, permissioning, and literacy on the access side from decision rights, override authority, and incentives on the agency side. Test dimensionality, reliability, and sensitivity to change. Link scores to independent outcomes such as appeal resolution quality, override precision, role autonomy, and retention.

Calibrate the Decision Compass. Build domain anchored rubrics for the five factors and pre register promotion and reversion thresholds. Compare composite rules with rules that use explicit ceilings and vetoes. Study interrater reliability and the impact of separating scoring authority from operational ownership. Evaluate whether alternative factors, such as societal externalities or concentration of power, improve predictive value.

Run field experiments on modes. Randomize tasks or teams to Assist versus Interface conditions with identical content and controls. Measure accuracy, timeliness, error stability, override quality, and downstream equity. Examine heterogeneous effects by experience and by task novelty. Identify when Interface increases cognitive load and when Assist induces overreliance.

Estimate residual effort with telemetry. Test procedures for turning suggested, accepted, edited, and rejected events into stable residual estimates. Compare human corrected error rates with fully human baselines to quantify true gains. Study how residuals change with training, prompt libraries, and documentation quality. Report headcount effects based on internal standards and track divergence between budget and throughput planning.

Assess role quality and human outcomes. Use validated work design and well being instruments to test whether rebundling increases task identity, autonomy, and feedback. Track learning, fairness perceptions, and appeal outcomes. Examine whether Human Only guardrails protect stewardship in high stakes decisions or simply relocate pressure to unrecorded channels.

Portfolio steering and dynamic capabilities. At the enterprise level, test whether measured movement toward empowerment on the AI Matrix correlates with improvements in cycle time stability, defect rates, and service reliability. Study cadence of steering cycles, common failure modes in mode transitions, and the governance load required to sustain them. Evaluate whether telemetry driven steering produces genuine ambidexterity or creates churn.

Equity, explainability, and due process. Stress test the documentation stack. Measure whether rationale capture improves decision quality or only traceability. Quantify how often appeals overturn decisions and whether outcomes differ across groups. Test the effectiveness of uncertainty displays and evidence disclosures in reducing automation bias. Where people are affected, treat equity screens as primary acceptance criteria, not optional diagnostics.

Vendor portability and provenance. Compare outcomes in closed and open toolchains with identical controls. Evaluate the cost and benefit of documentation depth, including model cards, data sheets, and audit trails, for different risk classes. Study how provenance signals and watermarks affect trust, error recovery, and compliance burden.

Contexts beyond shared services. Extend to safety critical and public sector domains with layered oversight and mandatory auditing. Examine emerging market settings where data quality is uneven and infrastructure is constrained. Investigate collective agency arrangements where decision rights are shared with professional bodies or community oversight.

Treat the AI-OS loop and Compass rubrics as preregistered artifacts for replication. Publish a minimal telemetry schema, Compass rubrics, and analysis code as replication packages with appropriate privacy protections. Encourage registered reports for mode-contrast experiments and pre analysis plans for portfolio steering studies. Over time, assemble a benchmark that compares mode policies, control sets, and outcomes across organizations.

Taken together, these lines of work convert the model from a disciplined operating grammar into a cumulative research program. The aim is to show when and how balanced growth in access and agency produces measurable gains in reliability and role quality while protecting human stewardship where it matters most.

8. Conclusion

This paper presents the AI-OS as a single, governed system for human-AI work. The AI Matrix supplies the first-order target and diagnostic, the 3A Model sets the strategic rules of engagement, the Decision Compass translates risk into permitted modes with proportionate controls, and OI/IHACC provides the orchestration pattern that turns tasks and evidence into audience ready outputs. Figure 1 summarizes the AI-OS stack and the Section 6 execution loop.

The result is a coherent operating grammar that makes the human-AI boundary visible and reversible, links local decisions to enterprise movement on access and agency, and supports audit, appeal, and learning through documentation and telemetry. The minimal interface is a live task register with factor scores and rationales, permitted modes, control sets, KPI templates, escalation paths, residual priors, and documentation pointers, plus role rebundling rules and mode transition criteria.

The research agenda follows directly from the system contract. Measure access and agency with validity, calibrate Compass rubrics and thresholds, run field contrasts on modes, and test whether

telemetry driven portfolio steering produces durable gains in reliability, role quality, and equity while preserving human stewardship where stakes, ambiguity, or irreversibility are high.

In short, the contribution is a practical bridge from theory to practice that is simple enough to implement and strong enough to audit. By centering access and agency, designing explicit human–machine modes, and running a disciplined, telemetry-driven method, organizations can capture safe efficiency, improve reliability, and raise role quality without abandoning human stewardship where it matters most. Adopting AI-OS turns AI from a set of tools into a transparent, auditable system of work.

References

Acemoglu, D. and Restrepo, P. (2018) Artificial Intelligence, Automation and Work. Working Paper No. 24196. National Bureau of Economic Research. https://doi.org/10.3386/w24196

Acemoglu, D., & Restrepo, P. (2019). Automation and new tasks: How technology displaces and reinstates labor. *Journal of Economic Perspectives, 33*(2), 3–30. https://doi.org/10.1257/jep.33.2.3

Ackoff, R. L. (1989). From data to wisdom. *Journal of Applied Systems Analysis*, 16, 3–9.

Amershi, S., Weld, D. S., Vorvoreanu, M., Fourney, A., Nushi, B., Collisson, P., Suh, J., Iqbal, S. T., Bennett, P. N., Inkpen, K., Teevan, J., Kikin-Gil, R., & Horvitz, E. (2019). Guidelines for human-AI interaction. *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems (CHI '19)*, 1–13. https://doi.org/10.1145/3290605.3300233

Autor, D. H. (2015). Why are there still so many jobs? The history and future of workplace automation. Journal of Economic Perspectives, 29(3), 3–30. https://doi.org/10.1257/jep.29.3.3

Autor, D. H., Levy, F., & Murnane, R. J. (2003). The skill content of recent technological change: An empirical exploration. *The Quarterly Journal of Economics, 118*(4), 1279–1333. https://doi.org/10.1162/003355303322552801

Barley, S. R. (1986). Technology as an occasion for structuring: Evidence from observations of CT scanners and the social order of radiology departments. *Administrative Science Quarterly, 31*(1), 78–108. https://doi.org/10.2307/2392767

Barocas, S., & Selbst, A. D. (2016). Big data's disparate impact. *California Law Review, 104*(3), 671–732. https://doi.org/10.15779/Z38BG31

Bender, E. M., Gebru, T., McMillan-Major, A., & Shmitchell, S. (2021). On the dangers of stochastic parrots: Can language models be too big? In *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency (FAccT '21)* (pp. 610–623). Association for Computing Machinery. https://doi.org/10.1145/3442188.3445922

Brynjolfsson, E., Li, D., & Raymond, L. R. (2025). Generative AI at work. *The Quarterly Journal of Economics, 140*(2), 889–942. https://doi.org/10.1093/qje/qjae044

Brynjolfsson, E., Mitchell, T., & Rock, D. (2018). What can machines learn, and what does it mean for occupations and the economy? *AEA Papers and Proceedings, 108*, 43–47. https://doi.org/10.1257/pandp.20181019

Frey, C. B., & Osborne, M. A. (2017). The future of employment: How susceptible are jobs to computerisation? Technological Forecasting and Social Change, 114, 254–280. https://doi.org/10.1016/j.techfore.2016.08.019

Galbraith, J. R. (1974). Organization design: An information processing view. *Interfaces, 4*(3), 28–36. https://doi.org/10.1287/inte.4.3.28

Gebru, T., Morgenstern, J., Vecchione, B., Wortman Vaughan, J., Wallach, H., Daumé III, H., & Crawford, K. (2021). Datasheets for datasets. *Communications of the ACM, 64*(12), 86–92. https://doi.org/10.1145/3458723

Gerlich, M. (2025). From Offloading to Engagement: An Experimental Study on Structured Prompting and Critical Reasoning with Generative AI. *Data*, *10*(11), 172. https://doi.org/10.3390/data10110172
Hackman, J. R., & Oldham, G. R. (1976). Motivation through the design of work: Test of a theory. *Organizational Behavior and Human Performance, 16*(2), 250–279. https://doi.org/10.1016/0030-5073(76)90016-7
Holzinger, A. (2016). Interactive machine learning for health informatics: When do we need the human-in-the-loop? *Brain Informatics, 3*, 119–131. https://doi.org/10.1007/s40708-016-0042-6
International Labour Organization. (2023). *Generative AI and jobs: A global analysis of potential effects on job quantity and quality* (ILO Working Paper No. 96).
International Labour Organization. (2025). *Generative AI and jobs: A refined global index of occupational exposure* (ILO Working Paper No. 140).
Kasparov, G., Greengard, M. (2017). *Deep thinking: Where machine intelligence ends and human creativity begins*. PublicAffairs.
Kellogg, K. C., Valentine, M. A., & Christin, A. (2020). Algorithms at work: The new contested terrain of control. *Academy of Management Annals, 14*(1), 366–410. https://doi.org/10.5465/annals.2018.0174
March, J. G. (1991). Exploration and exploitation in organizational learning. *Organization Science, 2*(1), 71–87. https://doi.org/10.1287/orsc.2.1.71
Mateescu, A., & Nguyen, A. (2019). *Explainer: Algorithmic management in the workplace*. Data & Society. https://datasociety.net/library/explainer-algorithmic-management-in-the-workplace/
Mitchell, M., Wu, S., Zaldivar, A., Barnes, P., Vasserman, L., Hutchinson, B., Spitzer, E., Raji, I. D., & Gebru, T. (2019). Model cards for model reporting. In *Proceedings of the 2019 Conference on Fairness, Accountability, and Transparency (FAT '*19) (pp. 220–229). ACM. https://doi.org/10.1145/3287560.3287596
Mollick, E. (2024). *Co-intelligence: Living and working with AI*. Penguin Random House.
Noy, S., & Zhang, W. (2023). Experimental evidence on the productivity effects of generative artificial intelligence. *Science, 381*(6654), 187–192. https://doi.org/10.1126/science.adh2586
O'Reilly, C. A., III, & Tushman, M. L. (2013). Organizational ambidexterity: Past, present, and future. *Academy of Management Perspectives, 27*(4), 324–338. https://doi.org/10.5465/amp.2013.0025
Orlikowski, W. J. (1992). The duality of technology: Rethinking the concept of technology in organizations. *Organization Science, 3*(3), 398–427. https://doi.org/10.1287/orsc.3.3.398
Pfeffer, J., & Sutton, R. I. (2006). Hard facts, dangerous half-truths, and total nonsense: Profiting from evidence-based management. Harvard Business Review Press.
Project Management Institute. (2021). *A guide to the project management body of knowledge (PMBOK® Guide)* (7th ed.). Project Management Institute. ISBN 978-1628256642.
Raji, I. D., Smart, A., White, R. N., Mitchell, M., Gebru, T., Hutchinson, B., Smith-Loud, J., Theron, D., & Barnes, P. (2020). Closing the AI accountability gap: Defining an end-to-end framework for internal algorithmic auditing. *Proceedings of the 2020 Conference on Fairness, Accountability, and Transparency (FAT '*20), 33–44. https://doi.org/10.1145/3351095.3372873
Risko, E. F., & Gilbert, S. J. (2016). Cognitive offloading. *Trends in Cognitive Sciences, 20*(9), 676–688. https://doi.org/10.1016/j.tics.2016.07.002
Rosenblat, A., & Stark, L. (2016). Algorithmic labor and information asymmetries: A case study of Uber's drivers. *International Journal of Communication, 10*, 3758–3784. https://ijoc.org/index.php/ijoc/article/view/4892
Rowley, J. (2007). The wisdom hierarchy: Representations of the DIKW hierarchy. Journal of Information Science, 33(2), 163–180. https://doi.org/10.1177/0165551506070706
Selbst, A. D., Boyd, D., Friedler, S. A., Venkatasubramanian, S., & Vertesi, J. (2019). Fairness and abstraction in sociotechnical systems. In *Proceedings of the 2019 Conference on Fairness, Accountability, and Transparency (FAT '*19) (pp. 59–68). ACM. https://doi.org/10.1145/3287560.3287598

Simpson, E. (2025a) *Flow Acceleration and the IHACC Model: Human-AI Co-Creation in Epistemology*. Social Science Research Network. http://dx.doi.org/10.2139/ssrn.5742245
Simpson, E. (2025b), *Orchestrated Intelligence: Rethinking Knowledge Work in the Age of AI*. Social Science Research Network. http://dx.doi.org/10.2139/ssrn.5746322
Simpson, E. (2025c) The AI Matrix: Empowerment or Dependency? A Conceptual Framework. Social Science Research Network. http://dx.doi.org/10.2139/ssrn.5228571
Sparrow, B., Liu, J., & Wegner, D. M. (2011). Google effects on memory: Cognitive consequences of having information at our fingertips. *Science, 333*(6043), 776–778. https://doi.org/10.1126/science.1207745
Taylor, F. W. (1911). The principles of scientific management. Harper & Brothers.
Teece, D. J., Peteraf, M. A., & Leih, S. (2016). Dynamic capabilities and organizational agility: Risk, uncertainty, and strategy in the innovation economy. *California Management Review, 58*(4), 13–35. https://doi.org/10.1525/cmr.2016.58.4.13

Appendix A: Checklist

Note: Full details of model weights and methods are available under license. Contact the author for access to the Pro Toolkit and annex materials.

Purpose. Quick path from Decision Compass scoring to governable practice. Use after scoring a task on the five factors: Stakes, Ambiguity, Reversibility, Data Sensitivity, Reputation Risk.

1) Score to mode
• Autopilot
• AI-Human Interface
• Assist
• Human Only
Ceilings: if any single factor is high block Autopilot and require Interface or Human Only.

2) Required controls by mode
Autopilot: input validation; exception queue with service levels; periodic sampling audits; drift monitoring.
KPIs: defect rate; cycle time; exception rate; drift alerts.

AI-Human Interface: uncertainty and evidence display; rationale capture; override logging; time-bound appeals.
KPIs: agreement with ranked options; override frequency and reasons; decision latency; equity screen if people-affecting.

Assist: verification checklist; evidence or source logging; adversarial review for consequential outputs.
KPIs: post-verification accuracy; edit distance; time to insight.

Human Only: structured deliberation and peer review; documented reasons and evidence considered; equity and due process checks.
KPIs: due process completion; stability across comparable cases; appeal outcomes.

3) Residuals and rebundling
Record residual human effort as an adjustable prior and update with telemetry. Cluster residual tasks by journey, cadence, and decision mode.

4) Capacity translation
Convert after AI hours to FTE based on internal standards. Reconcile to baseline volumes and durations.

5) Pilot and transition gates
Green light candidates with low composite risk and clear acceptance thresholds. Collect telemetry on accuracy, timeliness, overrides, and equity. Promote, hold, or revert modes based on thresholds and stability.

6) Documentation and appeals
Record factor scores, mode assignment, control set, KPIs, and escalation path in the task register. Use a decision journal for Interface and Human Only.

7) Portfolio steering and AI Matrix
Map functions by AI Matrix quadrant and target investments that raise access and agency together. Track movement toward empowerment quadrants.